五 基于MindSpore推理的量化实验

5.1 实验目的

本实验的目的是了解神经网络量化操作，能够独立实现`int8`量化操作，构建量化`VGG16`神经网络，并基于MindSpore框架实现量化推理，能够独立编写量化操作代码。

5.2 背景介绍

5.2.1.量化

量化即以较低的推理精度损失将连续取值（或者大量可能的离散取值）的浮点型模型权重或流经模型的张量数据定点近似（通常为INT8）为有限多个（或较少的）离散值的过程，它是以更少位数的数据类型用于近似表示32位有限范围浮点型数据的过程，而模型的输入输出依然是浮点型。这样的好处是可以减小模型尺寸大小，减少模型内存占用，加快模型推理速度，降低功耗等。

如上所述，与FP32类型相比，FP16、INT8、INT4等低精度数据表达类型所占用空间更小。使用低精度数据表达类型替换高精度数据表达类型，可以大幅降低存储空间和传输时间。而低比特的计算性能也更高，INT8相对比FP32的加速比可达到3倍甚至更高，对于相同的计算，功耗上也有明显优势。

当前业界量化方案主要分为两种：感知量化训练（Quantization Aware Training）和训练后量化（Post-training Quantization）

5.2.2.训练后量化

对于已经训练好的`float32`模型，通过训练后量化将其转为`int8`，不仅能减小模型大小，而且能显著提高推理性能。训练后量化分为两类：

1. 权重量化：对模型的权值进行量化，仅压缩模型大小，推理时仍然执行`float32`推理；
2. 全量化：对模型的权值、激活值等统一进行量化，推理时执行`int`运算，能提升模型推理速度、降低功耗。

5.2.3.矩阵运算的int8量化

r代表浮点实数，q代表量化后的定点整数。浮点数和整型之间的换算公式如下所示，其中S代表缩放系数，表示实数和整数之间的比例关系，$r_{max}, r_{min}, q_{max}, q_{min}$分布代表浮点实数及定点整数最大最小值，$Z$代表实数中的0经过量化对应的整数值，为了在矩阵padding的时候保证浮点数值的0和定点整数的Z完全等价，保证定点和浮点之间的表征能够一致。

$$r = S(q - Z)$$

$$q = round(\frac{r}{S} + Z)$$

$$S = \frac{r_{max} - r_{min}}{q_{max} - q_{min}}$$

$$Z = round(q_{max} - \frac{r_{max}}{S})$$

假设卷积的权重$Weight$为w，$bias$为b，输入为x，输出的激活值为a。由于卷积本质上就是矩阵运算，因此表示成：

$$a = \sum_{i}^{N} w_i x_i + b$$

由此得到量化的公式

$$S_a(q_a - Z_a) = \sum_i^N S_w(q_w - Z_w)S_x(q_x - Z_x) + S_b(q_b - Z_b)$$

$$q_a = \frac{S_wS_x}{S_a}\sum_i^N(q_w - Z_w)(q_x - Z_x) + \frac{S_b}{S_a}(q_b - Z_b) + Z_a$$

这里面非整数部分就只有$\frac{S_wS_x}{S_a}$、$\frac{S_b}{S_a}$，因此接下来把这部分变成定点运算，对于b由于$\sum_i^N S_w(q_w - Z_w)S_x(q_x - Z_x)$的结果通常会用`int32`的整数存储，因此$b$通常也量化到`int32`，这里可以直接用$S_wS_x$来代表$S_b$，由于$S_w$、$S_x$都是对应8个`bit`的缩放比例，因此$S_wS_x$最多就放缩到16个`bit`，用32`bit`来存放$b$完全足够，$Z_b$直接记为0。因此公式调整为

$$\begin{aligned} q_a &= \frac{S_wS_x}{S_a}(\sum_i^N(q_w - Z_w)(q_x - Z_x) + q_b) + Z_a \\ &= M(\sum_i^N q_wq_x - \sum_i^N q_wZ_x - \sum_i^N q_xZ_w + \sum_i^N Z_wZ_x + q_b) + Z_a \end{aligned}$$

其中$M = \frac{S_wS_x}{S_a}$，M通常为$(0,1)$之间的实数，因此可表示为$M = 2^{-n}M_0$，其中M_0是一个定点实数，这样可以通过M_0的bit位移操作实现$2^{-n}M_0$，这样整个过程就都在定点上计算了。

由于Z_w、q_w、Z_x、q_b都是可以事先计算的，因此$\sum_i^N q_wZ_x$、$\sum_i^N Z_wZ_x + q_b$也可以事先计算好，实际推理的时候只需要计算$\sum_i^N q_wq_x$、$\sum_i^N q_xZ_w$即可。

这里解释一下为什么可以用S_xS_w来代替S_b，Z_b可以直接记为0：首先S和Z只是充当r和q之间转换的桥梁，只要保证r经过S、Z变换后得到q，而这个q经过S和Z可以反变换得到r即可。假设$r \in [-1,1], q \in [0,255]$那么$S = \frac{2}{255}$、$Z = 128$，一个$[-1,1]$区间的实数，完全可以通过$S$和$Z$可以换算到$[0,255]$区间的整数。如果对$q$的范围限制到$[0,100]$，$r$和$q$依然可以转换，但会把信息都压缩到更小的$[0,100]$区间了。所以用$S_xS_w$来代替$S_b$,$Z_b$记为0。假设所有的$r \in [-1,1]$那么$S_xS_w = \frac{4}{2^{16}}$，而$b$是用32bit存储的，本来$S_b = \frac{2}{2^{32}}$，因此缩放系数砍掉了一半的信息量，会带来一定的精度损失，不过大部分情况下这点损失是可以忽略的。

5.3 实验环境

平台：Modelarts

操作系统：euler（aarch64）

软件环境：编程框架MindSpore1.7.1、异构计算架构CANN5.1，Python3.7.10

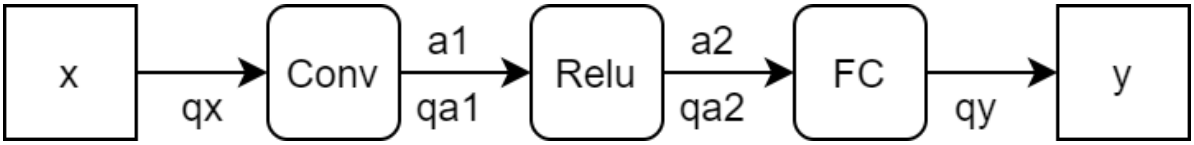
硬件环境：Ascend910A

测试数据集：花卉数据集（雏菊、蒲公英、玫瑰、向日葵、郁金香）

5.4 实验内容

5.5 实验步骤

对于一个简单的卷积网络，x、y代表输入和输出，a_1、a_2是网络中间的特征图，q_x、q_{a1}、q_{a2}、q_y表示量化后的定点数，在后训练量化中，需要一些样本来统计x、a_1、a_2、y的数值范围，在根据量化位数以及量化方法来计算S和Z。



1. 首先输入部分样本图片进行正向传播，获取输入，输出及中间特征图的最大最小值。其中对于Relu、Maxpool算子来说，会沿用上一层输出的min和max，不需要额外统计。
2. 根据min和max以及量化的位数，计算S和Z。
3. 在量化推理的时候，会把输入x量化成定点整数q_x，然后进行卷积计算，得到输出q_{a1}，这个结果依然是整型的，然后继续计算Relu的输出q_{a2}，对于FC也是矩阵运算，得到输出q_y，根据计算出来的S和Z推算回浮点整数y，除了输入输出的量化与反量化操作，其他流程完全可以用定点运算来完成。

5.5.1 量化模块

首先实现基本的量化公式，具体代码如下

$$S = \frac{r_{max} - r_{min}}{q_{max} - q_{min}}$$

$$Z = round(q_{max} - \frac{r_{max}}{S})$$

```
def calculate_scale_zero_point(min_val, max_val, num_bits=8):
    qmin = 0.
    qmax = 2. ** num_bits - 1.
    scale = float((max_val - min_val) / (qmax - qmin))  # S=(rmax-rmin)/(qmax-
qmin)
    zero_point = round(qmax - max_val / scale)  # Z=round(qmax-rmax/scale)

    if zero_point < qmin:
        zero_point = qmin
    elif zero_point > qmax:
        zero_point = qmax

    return scale, zero_point


def quantize_tensor(x, scale, zero_point, num_bits=8, signed=False):
    # TODO 请根据公式实现张量的量化操作
    if signed:
        qmin = - 2. ** (num_bits - 1)
        qmax = 2. ** (num_bits - 1) - 1
    else:
        qmin = 0.
        qmax = 2. ** num_bits - 1.

    q_x = zero_point + x / scale
    q_x = (q_x.clip(qmin, qmax)).round()  # q=round(r/S+Z)
    return q_x.astype(ms.float32)       # 由于mindspore不支持int类型的运算，因此我们还
是用float来表示整数

def dequantize_tensor(q_x, scale, zero_point):
```

```
    return scale * (q_x - zero_point)  # r=S(q-Z)
```

在量化过程中，需要先统计样本和中间层的最大最小值，同时也涉及到量化、反量化操作，因此将这些功能封装成一个QParam类

```
class QParam(object):
    def __init__(self, num_bits=8):
        self.num_bits = num_bits
        self.scale = None
        self.zero_point = None
        self.min = None
        self.max = None

    def update(self, tensor):
        # 用来统计 min、max
        if self.max is None or self.max < tensor.max():
            self.max = tensor.max()
        self.max = 0 if self.max < 0 else self.max

        if self.min is None or self.min > tensor.min():
            self.min = tensor.min()
        self.min = 0 if self.min > 0 else self.min

        self.scale, self.zero_point = calculate_scale_zero_point(self.min,
self.max, self.num_bits)

    def quantize_tensor(self, tensor):
        return quantize_tensor(tensor, self.scale, self.zero_point,
num_bits=self.num_bits)

    def dequantize_tensor(self, q_x):
        return dequantize_tensor(q_x, self.scale, self.zero_point)
```

接着定义基本的量化基类

- `__init__` 函数：指定量化的位数外，还需指定是否提供量化输入 (qi) 及输出参数 (qo)。在前面也提到，不是每一个网络模块都需要统计输入的 min、max，大部分中间层都是用上一层的 qo 来作为自己的 qi 的，另外有些中间层的激活函数也是直接用上一层的 qi 来作为自己的 qi 和 qo。
- `freeze` 函数：这个函数会在统计完 min、max 后发挥作用。正如上文所说的，公式 (4) 中有很多项是可以提前计算好的，freeze 就是把这些项提前固定下来，同时也将网络的权重由浮点实数转化为定点整数。
- `quantize_inference` 函数：这个函数主要是量化 inference 的时候会使用。

```
class QModule(nn.Cell):
    def __init__(self, qi=True, qo=True, num_bits=8):
        super(QModule, self).__init__()
        if qi:
            self.qi = QParam(num_bits=num_bits)
        if qo:
            self.qo = QParam(num_bits=num_bits)

    def freeze(self):
        pass

    def quantize_inference(self, x):
        raise NotImplementedError('quantize_inference should be implemented.')
```

5.5.2 量化卷积模块

量化卷积模块包括

- `__init__` 函数：需要传入 `conv_module` 模块，这个模块对应全精度的卷积层，另外的 `qw` 参数则是用来统计 weight 的 min、max 以及对 weight 进行量化用的。
- `freeze` 函数：这个函数主要就是计算公式中的 M、q_w、q_b，其中M应该由移位来实现定点化加速，为了实现方便，在此用原始的数学操作进行代替
- `construct` 函数：这个函数和正常的 construct一样，也是在 float 上进行的，只不过需要统计输入输出以及 weight 的 min、max 而已。其中这里需要对 weight 量化到 int8 然后又反量化回 float，这里其实就是所谓的伪量化节点，因为我们在实际量化 inference 的时候会把 weight 量化到 int8，这个过程本身是有精度损失的 (来自四舍五入的 round 带来的截断误差)，所以在统计 min、max 的时候，需要把这个过程带来的误差也模拟进去。
- `quantize_inference` 函数：这个函数在实际 inference 的时候会被调用。注意，这个函数里面的卷积操作是在 int 上进行的，这是量化推理加速的关键「当然，由于 mindspore的限制，我们仍然是在 float 上计算，只不过数值都是整数。这也可以看出量化推理是跟底层实现紧密结合的技术」。

```
class QConv2d(QModule):
    def __init__(self, conv_module, qi=True, qo=True, num_bits=8):
        super(QConv2d, self).__init__(qi=qi, qo=qo, num_bits=num_bits)
        self.num_bits = num_bits
        self.conv_module = conv_module
        self.qw = QParam(num_bits=num_bits)
        self.M = None

    def freeze(self, qi=None, qo=None):
        if hasattr(self, 'qi') and qi is not None:
            raise ValueError('qi has been provided in init function.')
        if not hasattr(self, 'qi') and qi is None:
            raise ValueError('qi is not existed, should be provided.')

        if hasattr(self, 'qo') and qo is not None:
            raise ValueError('qo has been provided in init function.')
        if not hasattr(self, 'qo') and qo is None:
            raise ValueError('qo is not existed, should be provided.')

        if qi is not None:
            self.qi = qi
        if qo is not None:
            self.qo = qo
```

```
        # TODO 请实现卷积模块权重参数量化
        self.M = self.qw.scale * self.qi.scale / self.qo.scale

        self.conv_module.weight =
self.qw.quantize_tensor(self.conv_module.weight)
        self.conv_module.weight = self.conv_module.weight - self.qw.zero_point

        self.conv_module.bias = quantize_tensor(self.conv_module.bias,
                                                scale=self.qi.scale *
self.qw.scale,
                                                zero_point=0, num_bits=32,
signed=True)

    def construct(self, x):
        if hasattr(self, 'qi'):
            self.qi.update(x)
            x = self.qi.quantize_tensor(x)
            x = self.qi.dequantize_tensor(x)

        self.qw.update(self.conv_module.weight)
        self.conv_module.weight =
self.qw.quantize_tensor(self.conv_module.weight)
        self.conv_module.weight =
self.qw.dequantize_tensor(self.conv_module.weight)
        x = ops.conv2d(x, self.conv_module.weight,
stride=self.conv_module.stride, pad_mode=self.conv_module.pad_mode)
        if self.conv_module.bias is not None:
            x = ops.bias_add(x, self.conv_module.bias)

        if hasattr(self, 'qo'):
            self.qo.update(x)
            x = self.qo.quantize_tensor(x)
            x = self.qo.dequantize_tensor(x)
        return x

    def quantize_inference(self, x):
        # TODO 请实现卷积模块量化推理模块
        x = x - self.qi.zero_point
        x = self.conv_module(x)
        x = self.M * x
        x = x.round()
        x = x + self.qo.zero_point
        x = x.clip(0., 2. ** self.num_bits - 1.).round()
        return x
```

5.5.3 量化全连接层模块

与量化卷积模块功能相似，这里不再叙述

```
class QDense(QModule):
    def __init__(self, fc_module, qi=True, qo=True, num_bits=8):
        super(QDense, self).__init__(qi=qi, qo=qo, num_bits=num_bits)
        self.num_bits = num_bits
        self.fc_module = fc_module
        self.qw = QParam(num_bits=num_bits)
        self.M = ms.Tensor([])
```

```
    def freeze(self, qi=None, qo=None):
        if hasattr(self, 'qi') and qi is not None:
            raise ValueError('qi has been provided in init function.')
        if not hasattr(self, 'qi') and qi is None:
            raise ValueError('qi is not existed, should be provided.')

        if hasattr(self, 'qo') and qo is not None:
            raise ValueError('qo has been provided in init function.')
        if not hasattr(self, 'qo') and qo is None:
            raise ValueError('qo is not existed, should be provided.')

        if qi is not None:
            self.qi = qi
        if qo is not None:
            self.qo = qo

        # TODO 请实现全连接模块权重参数量化
        self.M = self.qw.scale * self.qi.scale / self.qo.scale

        self.fc_module.weight = self.qw.quantize_tensor(self.fc_module.weight)
        self.fc_module.weight = self.fc_module.weight.data - self.qw.zero_point
        self.fc_module.bias = quantize_tensor(self.fc_module.bias,
                                              scale=self.qi.scale *
self.qw.scale,
                                              zero_point=0,
                                              num_bits=32,
                                              signed=True)

    def construct(self, x):
        if hasattr(self, 'qi'):
            self.qi.update(x)
            x = self.qi.quantize_tensor(x)
            x = self.qi.dequantize_tensor(x)

        self.qw.update(self.fc_module.weight)
        self.fc_module.weight = self.qw.quantize_tensor(self.fc_module.weight)
        self.fc_module.weight = self.qw.dequantize_tensor(self.fc_module.weight)
        x = ops.matmul(x, self.fc_module.weight.T)
        x = ops.bias_add(x, self.fc_module.bias)
        if hasattr(self, 'qo'):
            self.qo.update(x)
            x = self.qo.quantize_tensor(x)
            x = self.qo.dequantize_tensor(x)
        return x

    def quantize_inference(self, x):
        # TODO 请实现全连接模块量化推理
        x = x - self.qi.zero_point
        x = self.fc_module(x)
        x = self.M * x
        x = x.round()
        x = x + self.qo.zero_point
        x = x.clip(0., 2. ** self.num_bits - 1.).round()
        return x
```

5.5.4 量化ReLU模块

大体内容与量化卷积模块相似，其中需要注意，在`quantize_inference`函数中，量化零点非真实的0，需要特别注意。

```
class QReLU(QModule):
    def __init__(self, qi=False, num_bits=None):
        super(QReLU, self).__init__(qi=qi, num_bits=num_bits)

    def freeze(self, qi=None):
        if hasattr(self, 'qi') and qi is not None:
            raise ValueError('qi has been provided in init function.')
        if not hasattr(self, 'qi') and qi is None:
            raise ValueError('qi is not existed, should be provided.')
        if qi is not None:
            self.qi = qi

    def construct(self, x):
        if hasattr(self, 'qi'):
            self.qi.update(x)
            x = self.qi.quantize_tensor(x)
            x = self.qi.dequantize_tensor(x)
        x = ops.relu(x)

        return x

    def quantize_inference(self, x):
        x[x < self.qi.zero_point] = self.qi.zero_point
        return x
```

5.5.5 量化最大池化模块

大体内容与量化卷积模块相似，在量化推理时，因为最大池化原理就是取区域最大值作为输出，故直接进行算子运算即可。

```
class QMaxPooling2d(QModule):
    def __init__(self, max_pool_module, qi=False, num_bits=None):
        super(QMaxPooling2d, self).__init__(qi=qi, num_bits=num_bits)
        self.max_pool_module = max_pool_module

    def freeze(self, qi=None):
        if hasattr(self, 'qi') and qi is not None:
            raise ValueError('qi has been provided in init function.')
        if not hasattr(self, 'qi') and qi is None:
            raise ValueError('qi is not existed, should be provided.')
        if qi is not None:
            self.qi = qi

    def construct(self, x):
        if hasattr(self, 'qi'):
            self.qi.update(x)
            x = self.qi.quantize_tensor(x)
            x = self.qi.dequantize_tensor(x)
        x = self.max_pool_module(x)
        return x
```

```
    def quantize_inference(self, x):
        return self.max_pool_module(x)
```

5.5.6 量化VGG网络

量化卷积模块包括

- `__init__` 函数：基本的算子定义
- `construct` 函数：网络正向传播模块
- `quantize` 函数：量化网络模块
- `quantize_forward` 函数：量化正向传播模块
- `freeze` 函数：量化参数冻结模块
- `quantize_inference` 函数：量化推理模块

```
import mindspore.nn as nn
from quant_module import QConv2d, QMaxPooling2d, QDense, QReLU


class Vgg(nn.Cell):
    def __init__(self, num_classes=4):
        super(Vgg, self).__init__()
        self.layer1_conv1 = nn.Conv2d(in_channels=3, out_channels=64,
kernel_size=3, has_bias=True)
        self.layer1_relu1 = nn.ReLU()
        self.layer1_conv2 = nn.Conv2d(in_channels=64, out_channels=64,
kernel_size=3, has_bias=True)
        self.layer1_relu2 = nn.ReLU()
        self.layer1_maxpool = nn.MaxPool2d(kernel_size=2, stride=2)

        self.layer2_conv1 = nn.Conv2d(in_channels=64, out_channels=128,
kernel_size=3, has_bias=True)
        self.layer2_relu1 = nn.ReLU()
        self.layer2_conv2 = nn.Conv2d(in_channels=128, out_channels=128,
kernel_size=3, has_bias=True)
        self.layer2_relu2 = nn.ReLU()
        self.layer2_maxpool = nn.MaxPool2d(kernel_size=2, stride=2)

        self.layer3_conv1 = nn.Conv2d(in_channels=128, out_channels=256,
kernel_size=3, has_bias=True)
        self.layer3_relu1 = nn.ReLU()
        self.layer3_conv2 = nn.Conv2d(in_channels=256, out_channels=256,
kernel_size=3, has_bias=True)
        self.layer3_relu2 = nn.ReLU()
        self.layer3_conv3 = nn.Conv2d(in_channels=256, out_channels=256,
kernel_size=3, has_bias=True)
        self.layer3_relu3 = nn.ReLU()
        self.layer3_maxpool = nn.MaxPool2d(kernel_size=2, stride=2)

        self.layer4_conv1 = nn.Conv2d(in_channels=256, out_channels=512,
kernel_size=3, has_bias=True)
        self.layer4_relu1 = nn.ReLU()
        self.layer4_conv2 = nn.Conv2d(in_channels=512, out_channels=512,
kernel_size=3, has_bias=True)
        self.layer4_relu2 = nn.ReLU()
        self.layer4_conv3 = nn.Conv2d(in_channels=512, out_channels=512,
kernel_size=3, has_bias=True)
```

```
        self.layer4_relu3 = nn.ReLU()
        self.layer4_maxpool = nn.MaxPool2d(kernel_size=2, stride=2)

        self.layer5_conv1 = nn.Conv2d(in_channels=512, out_channels=512,
kernel_size=3, has_bias=True)
        self.layer5_relu1 = nn.ReLU()
        self.layer5_conv2 = nn.Conv2d(in_channels=512, out_channels=512,
kernel_size=3, has_bias=True)
        self.layer5_relu2 = nn.ReLU()
        self.layer5_conv3 = nn.Conv2d(in_channels=512, out_channels=512,
kernel_size=3, has_bias=True)
        self.layer5_relu3 = nn.ReLU()
        self.layer5_maxpool = nn.MaxPool2d(kernel_size=2, stride=2)
        self.flatten = nn.Flatten()

        self.fullyconnect1 = nn.Dense(512 * 7 * 7, 4096)
        self.relu_1 = nn.ReLU()
        self.fullyconnect2 = nn.Dense(4096, 4096)
        self.relu_2 = nn.ReLU()
        self.fullyconnect3 = nn.Dense(4096, num_classes)

    def construct(self, x):
        x = self.layer1_conv1(x)
        x = self.layer1_relu1(x)
        x = self.layer1_conv2(x)
        x = self.layer1_relu2(x)
        x = self.layer1_maxpool(x)

        x = self.layer2_conv1(x)
        x = self.layer2_relu1(x)
        x = self.layer2_conv2(x)
        x = self.layer2_relu2(x)
        x = self.layer2_maxpool(x)

        x = self.layer3_conv1(x)
        x = self.layer3_relu1(x)
        x = self.layer3_conv2(x)
        x = self.layer3_relu2(x)
        x = self.layer3_conv3(x)
        x = self.layer3_relu3(x)
        x = self.layer3_maxpool(x)

        x = self.layer4_conv1(x)
        x = self.layer4_relu1(x)
        x = self.layer4_conv2(x)
        x = self.layer4_relu2(x)
        x = self.layer4_conv3(x)
        x = self.layer4_relu3(x)
        x = self.layer4_maxpool(x)

        x = self.layer5_conv1(x)
        x = self.layer5_relu1(x)
        x = self.layer5_conv2(x)
        x = self.layer5_relu2(x)
        x = self.layer5_conv3(x)
        x = self.layer5_relu3(x)
        x = self.layer5_maxpool(x)
```

```
        x = self.flatten(x)
        x = self.fullyconnect1(x)
        x = self.relu_1(x)
        x = self.fullyconnect2(x)
        x = self.relu_2(x)
        x = self.fullyconnect3(x)
        return x

    def quantize(self, num_bits=8):
        # 第一个卷积模块需要获取量化输入，其余模块会复用之前的量化输出
        self.qlayer1_conv1 = QConv2d(self.layer1_conv1, qi=True, qo=True,
num_bits=num_bits)
        self.qlayer1_relu1 = QReLU()
        self.qlayer1_conv2 = QConv2d(self.layer1_conv2, qi=False, qo=True,
num_bits=num_bits)
        self.qlayer1_relu2 = QReLU()
        self.qlayer1_maxpool2d = QMaxPooling2d(self.layer1_maxpool)

        self.qlayer2_conv1 = QConv2d(self.layer2_conv1, qi=False, qo=True,
num_bits=num_bits)
        self.qlayer2_relu1 = QReLU()
        self.qlayer2_conv2 = QConv2d(self.layer2_conv2, qi=False, qo=True,
num_bits=num_bits)
        self.qlayer2_relu2 = QReLU()
        self.qlayer2_maxpool2d = QMaxPooling2d(self.layer2_maxpool)

        self.qlayer3_conv1 = QConv2d(self.layer3_conv1, qi=False, qo=True,
num_bits=num_bits)
        self.qlayer3_relu1 = QReLU()
        self.qlayer3_conv2 = QConv2d(self.layer3_conv2, qi=False,
qo=True,num_bits=num_bits)
        self.qlayer3_relu2 = QReLU()
        self.qlayer3_conv3 = QConv2d(self.layer3_conv3, qi=False, qo=True,
num_bits=num_bits)
        self.qlayer3_relu3 = QReLU()
        self.qlayer3_maxpool2d = QMaxPooling2d(self.layer3_maxpool)

        self.qlayer4_conv1 = QConv2d(self.layer4_conv1, qi=False, qo=True,
num_bits=num_bits)
        self.qlayer4_relu1 = QReLU()
        self.qlayer4_conv2 = QConv2d(self.layer4_conv2, qi=False, qo=True,
num_bits=num_bits)
        self.qlayer4_relu2 = QReLU()
        self.qlayer4_conv3 = QConv2d(self.layer4_conv3, qi=False, qo=True,
num_bits=num_bits)
        self.qlayer4_relu3 = QReLU()
        self.qlayer4_maxpool2d = QMaxPooling2d(self.layer4_maxpool)

        self.qlayer5_conv1 = QConv2d(self.layer5_conv1, qi=False, qo=True,
num_bits=num_bits)
        self.qlayer5_relu1 = QReLU()
        self.qlayer5_conv2 = QConv2d(self.layer5_conv2, qi=False, qo=True,
num_bits=num_bits)
        self.qlayer5_relu2 = QReLU()
        self.qlayer5_conv3 = QConv2d(self.layer5_conv3, qi=False, qo=True,
num_bits=num_bits)
        self.qlayer5_relu3 = QReLU()
        self.qlayer5_maxpool2d = QMaxPooling2d(self.layer5_maxpool)
```

```
        self.qfc1 = QDense(self.fullyconnect1, qi=False, qo=True,
num_bits=num_bits)
        self.qfc1_relu = QReLU()
        self.qfc2 = QDense(self.fullyconnect2, qi=False, qo=True,
num_bits=num_bits)
        self.qfc2_relu = QReLU()
        self.qfc3 = QDense(self.fullyconnect3, qi=False, qo=True,
num_bits=num_bits)

    def quantize_forward(self, x):
        x = self.qlayer1_conv1(x)
        x = self.qlayer1_relu1(x)
        x = self.qlayer1_conv2(x)
        x = self.qlayer1_relu2(x)
        x = self.qlayer1_maxpool2d(x)

        x = self.qlayer2_conv1(x)
        x = self.qlayer2_relu1(x)
        x = self.qlayer2_conv2(x)
        x = self.qlayer2_relu2(x)
        x = self.qlayer2_maxpool2d(x)

        x = self.qlayer3_conv1(x)
        x = self.qlayer3_relu1(x)
        x = self.qlayer3_conv2(x)
        x = self.qlayer3_relu2(x)
        x = self.qlayer3_conv3(x)
        x = self.qlayer3_relu3(x)
        x = self.qlayer3_maxpool2d(x)

        x = self.qlayer4_conv1(x)
        x = self.qlayer4_relu1(x)
        x = self.qlayer4_conv2(x)
        x = self.qlayer4_relu2(x)
        x = self.qlayer4_conv3(x)
        x = self.qlayer4_relu3(x)
        x = self.qlayer4_maxpool2d(x)

        x = self.qlayer5_conv1(x)
        x = self.qlayer5_relu1(x)
        x = self.qlayer5_conv2(x)
        x = self.qlayer5_relu2(x)
        x = self.qlayer5_conv3(x)
        x = self.qlayer5_relu3(x)
        x = self.qlayer5_maxpool2d(x)

        x = self.flatten(x)
        x = self.qfc1(x)
        x = self.qfc1_relu(x)
        x = self.qfc2(x)
        x = self.qfc2_relu(x)
        x = self.qfc3(x)
        return x

    def freeze(self):
        # 冻结网络参数时，除第一个卷积模块不需要指定量化输入，其余模块都需要指定量化输入
        self.qlayer1_conv1.freeze()
```

```
        self.qlayer1_relu1.freeze(qi=self.qlayer1_conv1.qo)
        self.qlayer1_conv2.freeze(qi=self.qlayer1_conv1.qo)
        self.qlayer1_relu2.freeze(qi=self.qlayer1_conv2.qo)
        self.qlayer1_maxpool2d.freeze(qi=self.qlayer1_conv2.qo)

        self.qlayer2_conv1.freeze(qi=self.qlayer1_conv2.qo)
        self.qlayer2_relu1.freeze(qi=self.qlayer2_conv1.qo)
        self.qlayer2_conv2.freeze(qi=self.qlayer2_conv1.qo)
        self.qlayer2_relu2.freeze(qi=self.qlayer2_conv2.qo)
        self.qlayer2_maxpool2d.freeze(qi=self.qlayer2_conv2.qo)

        self.qlayer3_conv1.freeze(qi=self.qlayer2_conv2.qo)
        self.qlayer3_relu1.freeze(qi=self.qlayer3_conv1.qo)
        self.qlayer3_conv2.freeze(qi=self.qlayer3_conv1.qo)
        self.qlayer3_relu2.freeze(qi=self.qlayer3_conv2.qo)
        self.qlayer3_conv3.freeze(qi=self.qlayer3_conv2.qo)
        self.qlayer3_relu3.freeze(qi=self.qlayer3_conv3.qo)
        self.qlayer3_maxpool2d.freeze(qi=self.qlayer3_conv3.qo)

        self.qlayer4_conv1.freeze(qi=self.qlayer3_conv3.qo)
        self.qlayer4_relu1.freeze(qi=self.qlayer4_conv1.qo)
        self.qlayer4_conv2.freeze(qi=self.qlayer4_conv1.qo)
        self.qlayer4_relu2.freeze(qi=self.qlayer4_conv2.qo)
        self.qlayer4_conv3.freeze(qi=self.qlayer4_conv2.qo)
        self.qlayer4_relu3.freeze(qi=self.qlayer4_conv3.qo)
        self.qlayer4_maxpool2d.freeze(qi=self.qlayer4_conv3.qo)

        self.qlayer5_conv1.freeze(qi=self.qlayer4_conv3.qo)
        self.qlayer5_relu1.freeze(qi=self.qlayer5_conv1.qo)
        self.qlayer5_conv2.freeze(qi=self.qlayer5_conv1.qo)
        self.qlayer5_relu2.freeze(qi=self.qlayer5_conv2.qo)
        self.qlayer5_conv3.freeze(qi=self.qlayer5_conv2.qo)
        self.qlayer5_relu3.freeze(qi=self.qlayer5_conv3.qo)
        self.qlayer5_maxpool2d.freeze(qi=self.qlayer5_conv3.qo)

        self.qfc1.freeze(qi=self.qlayer5_conv3.qo)
        self.qfc1_relu.freeze(qi=self.qfc1.qo)
        self.qfc2.freeze(qi=self.qfc1.qo)
        self.qfc2_relu.freeze(qi=self.qfc2.qo)
        self.qfc3.freeze(qi=self.qfc2.qo)

    def quantize_inference(self, x):
        # 对输入x进行量化
        qx = self.qlayer1_conv1.qi.quantize_tensor(x)

        qx = self.qlayer1_conv1.quantize_inference(qx)
        qx = self.qlayer1_relu1.quantize_inference(qx)
        qx = self.qlayer1_conv2.quantize_inference(qx)
        qx = self.qlayer1_relu2.quantize_inference(qx)
        qx = self.qlayer1_maxpool2d.quantize_inference(qx)

        qx = self.qlayer2_conv1.quantize_inference(qx)
        qx = self.qlayer2_relu1.quantize_inference(qx)
        qx = self.qlayer2_conv2.quantize_inference(qx)
        qx = self.qlayer2_relu2.quantize_inference(qx)
        qx = self.qlayer2_maxpool2d.quantize_inference(qx)

        qx = self.qlayer3_conv1.quantize_inference(qx)
```

```
        qx = self.qlayer3_relu1.quantize_inference(qx)
        qx = self.qlayer3_conv2.quantize_inference(qx)
        qx = self.qlayer3_relu2.quantize_inference(qx)
        qx = self.qlayer3_conv3.quantize_inference(qx)
        qx = self.qlayer3_relu3.quantize_inference(qx)
        qx = self.qlayer3_maxpool2d.quantize_inference(qx)

        qx = self.qlayer4_conv1.quantize_inference(qx)
        qx = self.qlayer4_relu1.quantize_inference(qx)
        qx = self.qlayer4_conv2.quantize_inference(qx)
        qx = self.qlayer4_relu2.quantize_inference(qx)
        qx = self.qlayer4_conv3.quantize_inference(qx)
        qx = self.qlayer4_relu3.quantize_inference(qx)
        qx = self.qlayer4_maxpool2d.quantize_inference(qx)

        qx = self.qlayer5_conv1.quantize_inference(qx)
        qx = self.qlayer5_relu1.quantize_inference(qx)
        qx = self.qlayer5_conv2.quantize_inference(qx)
        qx = self.qlayer5_relu2.quantize_inference(qx)
        qx = self.qlayer5_conv3.quantize_inference(qx)
        qx = self.qlayer5_relu3.quantize_inference(qx)
        qx = self.qlayer5_maxpool2d.quantize_inference(qx)

        qx = self.flatten(qx)

        qx = self.qfc1.quantize_inference(qx)
        qx = self.qfc1_relu.quantize_inference(qx)
        qx = self.qfc2.quantize_inference(qx)
        qx = self.qfc2_relu.quantize_inference(qx)
        qx = self.qfc3.quantize_inference(qx)

        # 对输出qx进行反量化
        out = self.qfc3.qo.dequantize_tensor(qx)
        return out
```

5.5.7 实验流程

第一步：初始化VGG网络并加载权重系数

第二步：构建对应推理数据

第三步：首先进行正常的网络推理，获取模型输出

第四步：构建量化模型，此实验为`int8`量化

第五步：进行量化推理，这里涉及到对中间特征图统计最大最小值

第六步：对网络量化参数进行固定

第七步：进行量化推理

```
import numpy as np
import cv2
import mindspore as ms
from mindspore import ops
from mindspore import load_checkpoint, load_param_into_net
from mindspore import context
from vgg import Vgg
```

```python
context.set_context(mode=context.PYNATIVE_MODE, device_target='CPU')
np.set_printoptions(suppress=True)


def resize_image(image, target_size):
    h, w = image.shape[:2]
    th, tw = target_size
    # 获取等比缩放后的尺寸
    scale = min(th / h, tw / w)
    oh, ow = round(h * scale), round(w * scale)
    # 缩放图片，opencv缩放传入尺寸为（宽，高），这里采用线性差值算法
    image = cv2.resize(image, (ow, oh),
interpolation=cv2.INTER_LINEAR).astype(np.uint8)
    # 将剩余部分进行填充
    new_image = np.ones((th, tw, 3), dtype=np.uint8) * 114
    new_image[:oh, :ow, :] = image
    return new_image


def process_image(img_path):
    # 读取图片，opencv读图后格式是BGR格式，需要转为RGB格式
    image = cv2.imread(img_path, cv2.IMREAD_COLOR)
    image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    # 将图片等比resize至(224x224)
    image = resize_image(image, (224, 224))
    image = np.array(image, dtype=np.float32)
    # 将图片标准化
    image -= [125.307, 122.961, 113.8575]
    image /= [51.5865, 50.847, 51.255]
    # (h,w,c) -> (c,h,w) -> (1,c,h,w)
    image = image.transpose((2, 0, 1))[None]
    return image


def direct_quantize(model, dataset):
    print('*'*50)
    print('Start quantize')
    for img_path, label in dataset:
        print("Start inference: {}".format(img_path))
        ndarray = process_image(img_path)
        tensor = ms.Tensor(ndarray, ms.float32)
        net_out = model.quantize_forward(tensor)
        prob = ops.Softmax()(net_out)
        print('Predict probability: {}'.format(np.around(prob.asnumpy(), 4)))
        predict_cls = (ops.Argmax()(prob)).asnumpy().item()
        print('Inference result: {}\n'.format(predict_cls == label))


def full_inference(model, dataset):
    print('*' * 50)
    print('Start full inference')
    for img_path, label in dataset:
        print("Start inference: {}".format(img_path))
        ndarray = process_image(img_path)
        tensor = ms.Tensor(ndarray, ms.float32)
        net_out = model(tensor)
        prob = ops.Softmax()(net_out)
        print('Predict probability: {}'.format(np.around(prob.asnumpy(), 4)))
```

```
        predict_cls = (ops.Argmax()(prob)).asnumpy().item()
        print('Inference result: {}\n'.format(predict_cls == label))


def quantize_inference(model, dataset):
    print('*' * 50)
    print('Start quantize inference')
    for img_path, label in dataset:
        print("Start inference: {}".format(img_path))
        ndarray = process_image(img_path)
        tensor = ms.Tensor(ndarray, ms.float32)
        net_out = model.quantize_inference(tensor)
        prob = ops.Softmax()(net_out)
        print('Predict probability: {}'.format(np.around(prob.asnumpy(), 4)))
        predict_cls = (ops.Argmax()(prob)).asnumpy().item()
        print('Inference result: {}\n'.format(predict_cls == label))


if __name__ == '__main__':
    # 初始化VGG网络并加载权重系数
    net = Vgg(num_classes=4)
    load_param_into_net(net, load_checkpoint('vgg.ckpt'), strict_load=True)
    net.set_train(False)

    # 构建对应推理数据
    dataset = [('./data/daisy_demo.jpg', 0),
               ('./data/roses_demo.jpg', 1),
               ('./data/sunflowers_demo.jpg', 2),
               ('./data/tulips_demo.jpg', 3)]

    # 首先进行正常的网络推理，获取模型输出
    full_inference(net, dataset)

    # 构建量化模型，此实验为int8量化
    net.quantize(num_bits=8)

    # 进行量化推理，这里涉及到对中间特征图统计最大最小值
    direct_quantize(net, dataset)

    # 对网络量化参数进行固定
    net.freeze()

    # 进行量化推理
    quantize_inference(net, dataset)
```

5.5.8 实验运行

代码目录介绍

```
EXP
|- data
|    |- daisy_demo.jpg            # 测试图片
|    |- roses_demo.jpg            # 测试图片
|    |- sunflowers_demo.jpg       # 测试图片
|    |- daisy_demo.jpg            # 测试图片
|- quant_inference.py             # 量化推理主函数
|- quant_module.py                # 量化模块
|- vgg.ckpt                       # VGG网络权重
|- vgg.py                         # VGG网络模块
```

1. 实现代码

```
vim quant_module.py
vim vgg.py
```

2. 运行实验

```
python3.7 quant_inference.py
```

5.6 评分指标

评估指标：

80：成功构建量化VGG网络，并输出正确结果

90：完成对量化VGG网络参数固定

100：完成对量化VGG网络进行量化推理，并输出正确结果

5.7 实验思考

在推理的时候，为了实现网络加速优化，通常将Convolution算子、BatchNormalization算子、Relu算子进行算子融合。若不进行融合，请思考如何实现BatchNormalization的量化模块。